KEDGE Business School EBP1 Quantitative Methods 1

# Worksheet 2 — Measures of central tendency

**Fully worked solutions**

Badr ABOUFARASSE, PhD — 2026–2027

## 1 Core applications

Master the essential definitions and calculations from the course.

*Core application — 25 min*

### 1.1 Exercise 1 – Choosing and calculating the appropriate measure of centre

1. Basket values are $18, 20, 20, 22, 25, 95$ euros. Calculate mode, median and arithmetic mean. Which measure best represents a typical customer? Why can the mean still matter?
2. Three grades are 14, 11 and 16, weighted by 30%, 30% and 40%. Calculate the weighted mean.
3. Revenue changes by $+5\%$, $-2\%$ and $+12\%$. Calculate the appropriate average annual growth rate.
4. A vehicle travels 120 km at 60 km/h and then 120 km at 90 km/h. Calculate the appropriate average speed.

**Fully worked solution**

**A. Mode, median and mean**

$$Mo = 20, \qquad Me = \frac{20+22}{2} = 21,$$

$$\bar{x} = \frac{18+20+20+22+25+95}{6} = 33.33.$$

The median better represents a typical customer because 95 is extreme. The mean is still useful for revenue because $n\bar{x}$ reproduces the total basket value.

**B. Weighted mean**

$$\bar{x}_w = 14(0.30) + 11(0.30) + 16(0.40) = 13.90.$$

**C. Geometric mean**

$$1 + g = \sqrt[3]{1.05 \times 0.98 \times 1.12} \approx 1.048442,$$

$$g \approx 4.84\%.$$

**D. Harmonic mean**

$$H = \frac{2}{\frac{1}{60} + \frac{1}{90}} = 72 \text{ km/h}.$$

*Core application — 10 min*

### 1.2 Exercise 3 – Which indicator should be used?

For each situation, choose the most appropriate indicator and justify in one sentence: most frequently sold product category; typical salary with a few very highly paid executives; average price when purchased quantities differ; average multi-year growth; average speed over equal distances; centre of a roughly symmetric numerical distribution without extreme values.

**Fully worked solution**

| Situation | Preferred indicator |
|---|---|
| Most frequently sold category | Mode. |
| Typical salary with a few extreme high salaries | Median. |
| Average price with different quantities | Weighted mean. |
| Average growth over several years | Geometric mean. |
| Average speed over equal distances | Harmonic mean. |
| Symmetric numerical distribution without extremes | Arithmetic mean. |

*Core application — 15 min*

### 1.3 Extra 1 — Robust centre for delivery performance

Delivery times (minutes) for 12 orders are: 18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 65.

1. Calculate the mode, median and mean.
2. Recalculate the mean without the 65-minute delivery.
3. Recommend the principal indicator for a customer-facing dashboard and justify your choice.

**Method hint:**

Compare the indicators before and after removing the unusual observation; do not delete it without explanation.

**Fully worked solution**

**Step 1 — mode.** The values 20, 21 and 22 each occur twice, so the distribution is multimodal.

**Step 2 — median.** With 12 observations, use positions 6 and 7: $(21 + 22)/2 = \mathbf{21.5\ minutes}$.

**Step 3 — mean.** The total is 300, hence the mean is $300/12 = \mathbf{25\ minutes}$. Without the unusual delivery, the total is 235 for 11 orders and the mean is $235/11 = \mathbf{21.36\ minutes}$.

**Interpretation.** A single long delivery raises the mean by 3.64 minutes. The median is the best principal description of a typical customer experience, but the dashboard should also report the mean and the 65-minute incident because the latter matters for capacity and service recovery.

## 2 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

*Assessment / exam practice — 30 min*

### 2.1 Exercise 2 – Grouped distribution

| Class | $[10, 14[$ | $[14, 18[$ | $[18, 26[$ | $[26, 30[$ | $[30, 35[$ | $[35, 40[$ | $[40, 45[$ | $[45, 50[$ | $[50, 55[$ |
|---|---|---|---|---|---|---|---|---|---|
| $n_i$ | 4 | 12 | 32 | 32 | 60 | 35 | 25 | 5 | 5 |

1. Identify the modal class, taking unequal widths into account.

2. Calculate the median by linear interpolation.
3. Calculate $Q_1$ and $Q_3$ by interpolation.
4. Calculate the approximate mean using class midpoints.
5. Find the maximum price such that about 60% of customers can afford at least that price.

**Fully worked solution**

The count densities are $1, 3, 4, 8, 12, 7, 5, 1, 1$, so the modal class is $[30, 35[$.

$$Me = 30 + 5\frac{105 - 80}{60} \approx 32.08.$$

$$Q_1 = 26 + 4\frac{52.5 - 48}{32} \approx 26.56,$$

$$Q_3 = 35 + 5\frac{157.5 - 140}{35} = 37.50.$$

Class midpoints are $12, 16, 22, 28, 32.5, 37.5, 42.5, 47.5, 52.5$ and

$$\sum n_i c_i = 6665.$$

Hence:

$$\bar{x} = \frac{6665}{210} \approx 31.74.$$

For 60% to afford at least price $p$, 40% must lie below it:

$$0.40(210) = 84,$$

$$p = 30 + 5\frac{84 - 80}{60} \approx 30.33 \text{ euros.}$$

**Full table for interpolation**

| Class | n | a | d=n/a | ECC | c | n×c |
|---|---|---|---|---|---|---|
| [10,14[ | 4 | 4 | 1 | 4 | 12 | 48 |
| [14,18[ | 12 | 4 | 3 | 16 | 16 | 192 |
| [18,26[ | 32 | 8 | 4 | 48 | 22 | 704 |
| [26,30[ | 32 | 4 | 8 | 80 | 28 | 896 |
| [30,35[ | 60 | 5 | 12 | 140 | 32.5 | 1950 |
| [35,40[ | 35 | 5 | 7 | 175 | 37.5 | 1312.5 |
| [40,45[ | 25 | 5 | 5 | 200 | 42.5 | 1062.5 |
| [45,50[ | 5 | 5 | 1 | 205 | 47.5 | 237.5 |
| [50,55[ | 5 | 5 | 1 | 210 | 52.5 | 262.5 |
| Σ | 210 | — | — | — | — | 6665 |

$$Q_1 = 26 + 4\frac{52.5 - 48}{32} = 26.5625,$$
$$Me = 30 + 5\frac{105 - 80}{60} = 32.0833,$$
$$Q_3 = 35 + 5\frac{157.5 - 140}{35} = 37.5.$$

Ranks: 0.25×210=52.5; 0.50×210=105; 0.75×210=157.5. Interpolation is approximate because the underlying individual observations are unknown.

*Assessment / exam practice — 20 min*

### 2.2 Extra 2 — Integrated centre-selection case

A retailer reports three indicators:

| Question | Data |
|---|---|
| Average basket value | 120 baskets: €3,600 total |
| Typical annual growth | +12%, −8%, +15%, +5% |
| Average speed | Two equal 90-km legs at 60 and 90 km/h |

1. Choose and calculate the appropriate mean for each row.
2. Explain why using the arithmetic mean in all three cases is incorrect.
3. Write a short managerial conclusion.

**Method hint:**

Match additive totals, compounded factors and equal-distance rates with arithmetic, geometric and harmonic means.

**Fully worked solution**

**Basket value.** The arithmetic mean preserves the total: 3,600/120 = **€30**.

**Growth.** Multiply the growth factors: $1.12 \times 0.92 \times 1.15 \times 1.05 = 1.244208$. The equivalent annual factor is $1.244208^{1/4} = 1.0561$, hence **5.61% per year**.

**Speed.** Equal distances require the harmonic mean: $2/(1/60 + 1/90) =$ **72 km/h**.

**Why the methods differ.** Basket values combine additively, growth compounds multiplicatively, and equal-distance speeds imply unequal travel times. The arithmetic alternatives (6% growth and 75 km/h) do not preserve the relevant totals.

**Managerial conclusion.** The average basket is €30, the equivalent annual growth rate is about 5.6%, and the route's average speed is 72 km/h. Each result is valid because the averaging rule matches the business mechanism.